Minimax recovery of a Darcy conductivity near a constant

Abstract

Let u_f solve $\operatorname{div}(f\nabla u) = 1$ in the unit ball $D \subset \mathbb{R}^d$, $d \geq 2$, with zero Dirichlet data, and suppose u_f is observed in Gaussian white noise of level $n^{-1/2}$. We determine the minimax rate for recovering the conductivity f in squared $L^2(D)$ loss over a small Hölder–Zygmund ball $\{\|f-1\|_{C^\alpha(D)} \leq r\}$, $\alpha > 1$: it is $n^{-2\alpha/(2\alpha+d+3)}$, with no logarithmic loss. The upper bound inverts the transport equation $\nabla u \cdot \nabla f + (\Delta u) f = 1$ along the gradient flow of the smoothed data; a bias cancellation for smoothed products and a variance gain of one power of the bandwidth from integrating the differentiated noise along characteristics drive the rate. The lower bound prescribes exact solutions oscillating in the $d-1$ angular directions, recovers their conductivities through the same transport inverse, and applies Fano's inequality.

1 Introduction

1.1 The problem

Let $D = \{x \in \mathbb{R}^d : |x| < 1\}$ be the open unit ball, $d \geq 2$, and write $B_\rho = \{x : |x| < \rho\}$. A *conductivity* is a bounded measurable function f on D bounded below by a positive constant. For such f, Darcy's problem

$$\operatorname{div}(f\nabla u) = 1 \ \text{ in } D, \qquad u = 0 \ \text{ on } \partial D, \tag{1.1}$$

has a unique weak solution $u_f \in H^1_0(D)$, by the Lax–Milgram theorem. The conductivity is unknown, and u_f is observed in Gaussian white noise of level $n^{-1/2}$:

$$Y_n = u_f + n^{-1/2}\mathbb{W}. \tag{1.2}$$

Concretely, for every $\varphi \in L^2(D)$ one observes $Y_n(\varphi) = \langle u_f, \varphi\rangle + n^{-1/2}\mathbb{W}(\varphi)$, where $\mathbb{W}$ is the isonormal Gaussian process on $L^2(D)$: the variables $\mathbb{W}(\varphi)$, $\varphi \in L^2(D)$, are jointly centred Gaussian with covariance $\mathbb{E}\,\mathbb{W}(\varphi)\mathbb{W}(\psi) = \langle \varphi, \psi\rangle$. We write P_f for the law of Y_n and $\mathbb{E}_f$ for the corresponding expectation. An *estimator* $\widehat{f}_n$ is any measurable function of the observed process with values in $L^2(D)$.

Hölder–Zygmund spaces. For $s > 0$ let $C^s(\mathbb{R}^d) = B^s_{\infty,\infty}(\mathbb{R}^d)$, with norm $\|F\|_{C^s} = \sup_{i\geq 0} 2^{si}\,\|F_i\|_\infty$, where $F = \sum_{i\geq 0} F_i$ is the inhomogeneous Littlewood–Paley decomposition (F_i frequency-localized to $\{2^{i-1} \leq |\xi| \leq 2^{i+1}\}$ for $i \geq 1$) [3, Section 4.3], [11]. For noninteger s this is the classical Hölder space; for integer s it is the Zygmund class, strictly larger than the space of functions with bounded continuous derivatives of order s. The space $C^s(D)$ consists of the restrictions to D of elements of $C^s(\mathbb{R}^d)$, with the quotient norm $\|f\|_{C^s(D)} = \inf\{\|F\|_{C^s(\mathbb{R}^d)} : F|_D = f\}$ [3, Section 4.3.6], normalized so that $\|f\|_{L^\infty(D)} \leq \|f\|_{C^s(D)}$. Integer indices always refer to the Zygmund scale, and classical derivative bounds are written explicitly as $\left\|D^k F\right\|_\infty$. For an integer $k < s$, Bernstein's inequality $\left\|D^k F_i\right\|_\infty \leq C 2^{ki}\,\|F_i\|_\infty$ [3, Lemma 4.3.4] makes $\sum_i D^k F_i$ converge uniformly, so every $F \in C^s$ has bounded continuous derivatives of order k with $\left\|D^k F\right\|_\infty \leq C\,\|F\|_{C^s}$ (on D, apply this to a bounded extension). Thus, for $\alpha > 1$, $f \in C^\alpha(D)$ has a bounded gradient and $u \in C^{\alpha+1}(D)$ has bounded second derivatives.

The parameter class. For $\alpha > 1$ and $r \in (0,1)$ let $\mathcal{F}(\alpha, r)$ be the set of all $f \in C^\alpha(D)$ with $\|f - \mathbf{1}\|_{C^\alpha(D)} \leq r$, where $\mathbf{1}$ is the constant function 1. Its members satisfy $1 - r \leq f \leq 1 + r$, so they are conductivities. The question is the order, as $n \to \infty$ for small fixed r, of the minimax risk $\mathcal{R}_n(\alpha, r)$ defined in the following theorem.

1.2 Main result

Theorem 1.1. *For every $d \geq 2$ and $\alpha > 1$ there is $r_0 > 0$ such that, for each fixed $0 < r \leq r_0$, there are constants $0 < c < C < \infty$ with*

$$c\, n^{-\frac{2\alpha}{2\alpha+d+3}} \;\leq\; \mathcal{R}_n(\alpha, r) = \inf_{\widehat{f}_n} \sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \left\| \widehat{f}_n - f \right\|^2_{L^2(D)} \;\leq\; C\, n^{-\frac{2\alpha}{2\alpha+d+3}}$$

for all sufficiently large n.

The threshold r_0 depends on α, d only; c, C may depend on α, d, r. Smallness of r is used only in the upper bound, through Lemma 2.1, the perturbative description of u_f that makes it uniformly strictly convex (Hessian bounded below by a positive multiple of the identity); the lower bound only needs its alternatives to lie in $\mathcal{F}(\alpha, r)$.

1.3 The transport equation, the exponent, and the plan

Expanding the divergence in (1.1) shows that f solves a *first-order linear transport equation* whose coefficients are determined by the solution:

$$T_v q := \operatorname{div}(q \nabla v) = \nabla v \cdot \nabla q + (\Delta v)\, q, \qquad T_{u_f} f = 1. \tag{1.3}$$

For $f \equiv 1$ the solution is $u_*(x) = (|x|^2 - 1)/(2d)$, with $\nabla u_* = x/d$, $D^2 u_* = d^{-1} I$ and $\Delta u_* = 1$. For f in a small ball around 1, Lemma 2.1 shows that u_f is a small $C^{\alpha+1}$ perturbation of u_*, hence uniformly strictly convex with outward-pointing gradient on ∂D and a single interior

critical point. The characteristics of T_{u_f} are the gradient flow lines of u_f, which all emanate from that critical point, and along each of them, in the direction of increasing u_f, $T_{u_f} f = 1$ is a scalar linear ODE damped by the coefficient $\Delta u_f \geq 1/2$. Integrating it outward from the critical point gives an explicit bounded inverse T_v^{-1} (Lemma 2.3), so that $f = T_{u_f}^{-1} 1$ *without any boundary values of* f. Treating (1.1) as a hyperbolic equation for the coefficient goes back to Richter [9]; see [5] for gradient-flow characteristics in a statistical setting.

The exponent arises as follows. Smoothing the data at bandwidth h yields a kernel smoother $V_h \approx u_f$ (Section 3.1) whose second derivatives carry noise of pointwise size $n^{-1/2} h^{-d/2-2}$. A crude bound on the plug-in $T_{V_h}^{-1} 1$ gives a squared noise contribution $n^{-1} h^{-d-4}$, against a bias of order h^α (itself the result of a cancellation, Section 3.2), and balancing would give the denominator $2\alpha + d + 4$. The variance is in fact at most $n^{-1} h^{-d-3}$: T_v^{-1} integrates along characteristics, which averages out one power of h in the flow direction while leaving the $d-1$ transverse directions untouched (Lemma 2.4), and $h^{2\alpha} = n^{-1} h^{-d-3}$ gives $2\alpha + d + 3$. The lower bound matches: solutions $u_\theta = u_* + v_\theta$ oscillating at frequency h^{-1} in the $d-1$ angular directions only, with L^2 size $h^{\alpha+2}$, have conductivities $f_\theta - 1 \approx -T_{u_*}^{-1} \Delta v_\theta$ of L^2 size h^α; with $h^{-(d-1)}$ independent bumps, the Kullback–Leibler budget $n h^{2\alpha+4} \lesssim h^{-(d-1)}$ gives the same h.

Plan. Section 2 collects four tools: the perturbative description of u_f (Lemma 2.1); the *product-smoothing cancellation* (Lemma 2.2): for a kernel smoother S_h with many vanishing moments, $(S_h F)(S_h G) - S_h(FG)$ is of order $h^{\alpha+1}$, one order better than $S_h F - F = O(h^\alpha)$, also after up to two derivatives; the transport inverse T_v^{-1} for uniformly convex v (Lemma 2.3); and the *variance gain* $\mathbb{E}\|T_v^{-1} \operatorname{div}(a \nabla e)\|_2^2 \lesssim n^{-1} h^{-d-3}$ for deterministic v, a and smoothed noise e (Lemma 2.4). Section 3 proves the upper bound: the plug-in operator T_{V_h} is random, so no independence can be exploited; instead an exact identity (3.12) expresses the error through the *deterministic* inverse $T_{u_h}^{-1}$, u_h the smoothed true solution, applied to a linear noise term plus a quadratic remainder controlled on a high-probability event. Section 4 prescribes the solutions u_θ and recovers $f_\theta = T_{u_\theta}^{-1} 1$, so that the Kullback–Leibler divergences are exact Gaussian shifts; Fano's inequality on a Varshamov–Gilbert packing finishes the proof.

1.4 Related work, notation and conventions

The white-noise Darcy model (1.2) is studied in Nickl, van de Geer and Wang [8], who obtain, for integer $\alpha > \max\{d/2 + 2,\, 2d - 1\}$ and Sobolev-type classes with boundary restrictions on f, the prediction rate $n^{-(\alpha+1)/(2\alpha+d+2)}$ for $\mathbb{E}\|\widehat{u} - u_f\|_{L^2}$, $\widehat{u}$ their estimator of u_f, and a conductivity bound of the form $n^{-(\alpha-1)/(2\alpha+d+2)}$ for the expected (non-squared) L^2 error via a stability estimate [8, Theorems 9–11]; their lower-bound construction already uses the unit ball, the source 1, the baseline $f = 1$ and u_*. Posterior contraction for Darcy appears in [1] and in the monograph [6]; information-theoretic aspects of nonlinear inverse problems, including Darcy with a constant background and transport equations, are discussed in [7]. We are not aware of an earlier determination of the minimax rate over $\mathcal{F}(\alpha, r)$; as the classes and smoothness ranges differ, the comparison with the rates above is not literal. The

product-smoothing estimate, the variance gain and the solution-first alternatives are proved here directly.

Constants $C, c, c', \dots$ are positive and finite, may change from line to line, and may depend on α, d, r and on the fixed kernels and bumps chosen below, but never on n, h or the true conductivity; subscripts indicate additional dependence (C_A, C_k). "For large n" means for $n \geq n_0$ with n_0 depending on the same quantities; $A \asymp B$ means $cB \leq A \leq CB$. $D^k F$ is the k-th derivative of F (a symmetric k-tensor) and $\left\|D^k F\right\|_\infty$ the supremum of its Euclidean norm; ν is the outward unit normal of a ball; matrix inequalities are in the quadratic-form sense and I is the identity. $\|\cdot\|_\infty$ and $\|\cdot\|_2$ are the supremum and L^2 norms over the domain under discussion (D unless another set is indicated). For $0 < h < 1/2$ put $\Omega_h = B_{1-h}$. Polar coordinates are $x = \rho\omega$, $\rho = |x|$, $\omega \in \mathbb{S}^{d-1}$, and $\Delta_{\mathbb{S}^{d-1}}$ is the Laplace–Beltrami operator of the unit sphere. Finally, $L_n = \sqrt{\log n}$.

2 Analytic tools

2.1 Regularity near the constant conductivity

Lemma 2.1. *There are $r_0 > 0$ and $C < \infty$, depending only on α and d, such that for $0 < r \leq r_0$ and every $f \in \mathcal{F}(\alpha, r)$,*

$$\|u_f - u_*\|_{C^{\alpha+1}(D)} \leq Cr, \qquad \|f\|_{C^\alpha(D)} + \|\nabla u_f\|_{C^\alpha(D)} \leq C. \tag{2.1}$$

In particular u_f is classically twice continuously differentiable on $\overline{D}$ and is a classical solution of (1.1)*, and*

$$\sum_{j=0}^{2} \left\|D^j(u_f - u_*)\right\|_\infty \leq Cr, \qquad \|f\|_\infty + \|\nabla f\|_\infty + \sum_{j=0}^{2} \left\|D^j u_f\right\|_\infty \leq C. \tag{2.2}$$

Proof. Let Δ_D^{-1} denote the solution operator of the Dirichlet problem for the Laplacian on D. We use the bound

$$\left\|\Delta_D^{-1} \operatorname{div} F\right\|_{C^{\alpha+1}(D)} \leq C \|F\|_{C^\alpha(D)} \tag{2.3}$$

for vector fields F. For noninteger α, $\operatorname{div} F$ lies in the classical Hölder space $C^{\alpha-1}$, and (2.3) is global Schauder theory for the Dirichlet problem: existence and uniqueness in $C^{2,\gamma}(\overline{D})$ (classical functions with γ-Hölder second derivatives), the global a priori estimate and higher boundary regularity [2, Theorems 6.14, 6.6 and 6.19]. At integer α the same linear operator is bounded between the spaces with two neighbouring noninteger indices, and real interpolation of Hölder–Zygmund spaces on smooth domains [11, Sections 1.6, 4.5 and 5.1–5.2] gives (2.3) at the endpoint; the image consists of functions vanishing on ∂D in every case, since the operator is the same.

Put $a = f - 1$, so $\|a\|_{C^\alpha(D)} \leq r$, and $v = u_f - u_*$. Since $\Delta u_* = 1$, the equation (1.1) for $u_f = u_* + v$ reads $\Delta v = -\operatorname{div}(a\nabla u_* + a\nabla v)$ with $v = 0$ on ∂D, i.e. $v = -\Delta_D^{-1} \operatorname{div}(a\nabla u_* + a\nabla v)$. Since pointwise multiplication is bounded $C^\alpha \times C^\alpha \to C^\alpha$ for $\alpha > 0$ (the algebra

property), $\|a\nabla v\|_{C^\alpha} \le C\,\|a\|_{C^\alpha}\,\|v\|_{C^{\alpha+1}} \le Cr\,\|v\|_{C^{\alpha+1}}$, so by (2.3) the right-hand side is a contraction of the space of $C^{\alpha+1}(D)$ functions with zero boundary trace when $r \le r_0$ is small, and its fixed point satisfies $\|v\|_{C^{\alpha+1}} \le C\,\|a\nabla u_*\|_{C^\alpha} \le Cr$. Since $\alpha+1>2$, the fixed point has bounded continuous second derivatives, so it is a classical solution and coincides with the unique weak solution; this proves (2.1), and (2.2) follows from the embedding recalled in Section 1.1 ($\alpha>1$ and $\alpha+1>2$). □

2.2 Kernel smoothing and the product-smoothing cancellation

Fix an integer $L > \alpha+3$ and a real kernel $K \in C_c^\infty(B_1)$ with

$$\int K = 1, \qquad \int x^\beta K(x)\,dx = 0 \quad (1 \le |\beta| < L). \tag{2.4}$$

Such a kernel exists, e.g. a suitable finite linear combination of dilates of one smooth bump, and need not be nonnegative; for background on higher-order kernels see [12, Section 1.2.2]. Set $K_h(x) = h^{-d}K(x/h)$ and $S_hF = K_h * F$. Because $\operatorname{supp} K$ is a compact subset of the open ball B_1, for x in a neighbourhood of $\overline{\Omega_h}$ the function $K_h(x-\cdot)$ is supported inside D: on $\Omega_h = B_{1-h}$ the smoother only uses values in D, and it applies to the white noise, $S_h\mathbb{W}(x) = \mathbb{W}(K_h(x-\cdot))$, a centred Gaussian field with smooth sample paths. Three elementary facts are used repeatedly on Ω_h: derivatives commute with S_h, $D^kS_hF = S_hD^kF$ whenever F has bounded continuous classical derivatives through order k, so $\left\|D^kS_hF\right\|_\infty \le \|K\|_{L^1}\left\|D^kF\right\|_\infty$; placing derivatives on the kernel instead gives $\left\|D^kS_hF\right\|_\infty \le Ch^{-k}\left\|F\right\|_\infty$; and by (2.4), S_h reproduces polynomials of degree less than L, in particular $S_hu_* = u_*$.

Lemma 2.2 (Product-smoothing cancellation)**.** *For F, G in a bounded subset of $C^\alpha(D)$, $\alpha>1$, and $0<h<1/2$,*

$$\|S_hF - F\|_{L^\infty(\Omega_h)} \le Ch^\alpha, \tag{2.5}$$

$$\left\|D^k\{(S_hF)(S_hG) - S_h(FG)\}\right\|_{L^\infty(\Omega_h)} \le Ch^{\alpha+1-k}, \qquad k=0,1,2. \tag{2.6}$$

These statements include integer Hölder–Zygmund indices.

The point of (2.6) is the exponent $\alpha+1$: each factor is approximated only to order h^α, but $Q_h(F,G) := (S_hF)(S_hG) - S_h(FG)$ vanishes identically when either argument is constant, so only increments of both F and G enter.

Proof. Replace F, G by extensions to $\mathbb{R}^d$ with bounded C^α norms; the convolutions on Ω_h are unchanged. Use the inhomogeneous dyadic decompositions $F = \sum_{i\ge0} F_i$, $G = \sum_{j\ge0} G_j$ of Section 1.1, so that

$$\|F_i\|_\infty, \|G_i\|_\infty \le C2^{-\alpha i}, \qquad \left\|D^\ell F_i\right\|_\infty \le C_\ell 2^{\ell i}\,\|F_i\|_\infty,$$

and likewise for G_i (dyadic characterization and Bernstein inequality, [3, Section 4.3], [11]). Let $2^J \le h^{-1} < 2^{J+1}$; blocks with index at most J are "low frequency". Since Q_h is bilinear, we estimate $Q_h(F_i, G_j)$ for low–low pairs individually and treat the remaining pairs in bulk.

Low–low pairs. As $Q_h(F,G)=Q_h(G,F)$, it suffices to take $i\le j\le J$, and we claim

$$\left\|D^k Q_h(F_i,G_j)\right\|_\infty \le Ch^L 2^i 2^{(L-1+k)j}\left\|F_i\right\|_\infty\left\|G_j\right\|_\infty. \tag{2.7}$$

The factor 2^i, rather than 2^{Li} or $2^{(L-1)i}2^j$, is the cancellation. It is seen from the exact identity (expand the product and use $\int K=1$)

$$Q_h(F,G)(x)=-\frac12\iint K(y)K(z)[F(x-hy)-F(x-hz)][G(x-hy)-G(x-hz)]\,dy\,dz.$$

Fix x and regard $q(t)=Q_t(F_i,G_j)(x)$ as a function of the bandwidth $t\in[0,h]$. By (2.4) and Taylor expansion, $S_tP(x)=P(x)+O(t^L)$ for every smooth P, so $q^{(l)}(0)=0$ for $l<L$ and $|q(h)|\le h^L\sup_{t\le h}|q^{(L)}(t)|/L!$. Differentiate the identity L times in t by the Leibniz rule. A term in which both brackets are differentiated, $a\ge1$ times the first and $L-a\ge1$ times the second, is bounded by $C2^{ai}2^{(L-a)j}\left\|F_i\right\|_\infty\left\|G_j\right\|_\infty\le C2^i2^{(L-1)j}\left\|F_i\right\|_\infty\left\|G_j\right\|_\infty$ since $i\le j$. If the first bracket is undifferentiated, bound it by its first difference, $|F_i(x-ty)-F_i(x-tz)|\le Ct2^i\left\|F_i\right\|_\infty$, and the second by $C2^{Lj}\left\|G_j\right\|_\infty$; since $t2^j\le1$ the same bound results, and symmetrically if the second bracket is undifferentiated (then $2^{Li}\cdot t2^j\le 2^i2^{(L-1)j}$). Spatial derivatives D^k fall on F_i or G_j and cost at most 2^{kj} by Bernstein's inequality, which proves (2.7). Summing over $i\le j\le J$ and using $\alpha>1$ in the inner sum,

$$Ch^L\sum_{j\le J}2^{(L-1+k-\alpha)j}\sum_{i\le j}2^{(1-\alpha)i}\le Ch^L2^{(L-1+k-\alpha)J}\le Ch^{\alpha+1-k}.$$

Pairs involving a high frequency. Put $F_{\mathrm{lo}}=\sum_{i\le J}F_i$, $F_{\mathrm{hi}}=F-F_{\mathrm{lo}}$, and similarly $G_{\mathrm{lo}},G_{\mathrm{hi}}$. Summing the block bounds,

$$\begin{aligned}\left\|F_{\mathrm{hi}}\right\|_\infty\le Ch^\alpha,\quad \left\|D^\ell F_{\mathrm{lo}}\right\|_\infty\le Ch^{1-\ell}\quad(\ell\ge1),\\ \left\|D^\ell(S_h-I)F_{\mathrm{lo}}\right\|_\infty\le Ch^{\alpha-\ell}\qquad(0\le\ell\le2),\end{aligned}$$

and the same for G; the last line uses $L>\alpha+3$ and the block bound $\left\|D^\ell(S_h-I)F_i\right\|_\infty\le C2^{(\ell-\alpha)i}(h2^i)^L$. For the mixed term write

$$\begin{aligned}&Q_h(F_{\mathrm{lo}},G_{\mathrm{hi}})=((S_h-I)F_{\mathrm{lo}})\,S_hG_{\mathrm{hi}}-[S_h,F_{\mathrm{lo}}]G_{\mathrm{hi}},\\ &[S_h,F_{\mathrm{lo}}]G(x):=S_h(F_{\mathrm{lo}}G)(x)-F_{\mathrm{lo}}(x)\,S_hG(x)=\int K_h(x-y)\{F_{\mathrm{lo}}(y)-F_{\mathrm{lo}}(x)\}G(y)\,dy.\end{aligned}$$

The first product has k-th derivative $O(h^{2\alpha-k})$ for $k\le2$, by the displayed bounds and $\left\|D^bS_hG_{\mathrm{hi}}\right\|_\infty\le Ch^{-b}\left\|G_{\mathrm{hi}}\right\|_\infty$. Differentiating the commutator k times in x and using $|F_{\mathrm{lo}}(y)-F_{\mathrm{lo}}(x)|\le C|y-x|$ together with $\left\|D^2F_{\mathrm{lo}}\right\|_\infty\le Ch^{-1}$ bounds it by $Ch^{1-k}\left\|G_{\mathrm{hi}}\right\|_\infty\le Ch^{\alpha+1-k}$. The term $Q_h(F_{\mathrm{hi}},G_{\mathrm{lo}})$ is symmetric, and the high–high term is $O(h^{2\alpha-k})$ by differentiating the kernels. Since $2\alpha>\alpha+1$, this proves (2.6). Finally, (2.5) follows by summing $\left\|(S_h-I)F_i\right\|_\infty\le C\min\{1,(h2^i)^L\}2^{-\alpha i}$ over i. □

2.3 Inverting the transport equation

Lemma 2.3 (Transport inverse)**.** *Let Ω be a ball with radius in $[1/2, 1]$, and let v be smooth on a neighbourhood of $\overline{\Omega}$ with*

$$mI \le D^2 v \le MI \quad \text{on } \overline{\Omega}, \qquad \nabla v \cdot \nu > 0 \quad \text{on } \partial\Omega, \qquad m = \frac{1}{2d}, \quad M = \frac{3}{2d}. \tag{2.8}$$

Then v has a unique critical point x_v in Ω, its minimum. Write $c_v = \Delta v$, so $c_v \ge dm = c_0 := 1/2$. Let Φ_t be the backward gradient flow of v, $\dot\Phi_t = -\nabla v(\Phi_t)$, $\Phi_0(x) = x$. For smooth q on $\overline{\Omega}$, the unique bounded classical solution of $T_v p = q$ on Ω is

$$p(x) = T_v^{-1} q(x) = \int_0^\infty J_t(x)\, q(\Phi_t(x))\, dt, \qquad J_t(x) = \exp\Big\{ -\int_0^t c_v(\Phi_s(x))\, ds \Big\}. \tag{2.9}$$

This solution is smooth, T_v^{-1} extends to a bounded operator on $L^2(\Omega)$, and

$$\left\| T_v^{-1} q \right\|_\infty \le c_0^{-1} \left\| q \right\|_\infty, \qquad \left\| T_v^{-1} q \right\|_2 \le 2c_0^{-1} \left\| q \right\|_2, \tag{2.10}$$

$$\left\| \nabla p \right\|_\infty \le (c_0 + m)^{-1} \big(\left\| \nabla q \right\|_\infty + \left\| \nabla c_v \right\|_\infty \left\| p \right\|_\infty \big). \tag{2.11}$$

Proof. Strict convexity and the outward boundary drift give a unique interior minimum x_v of v on $\overline{\Omega}$; the backward flow points into Ω at the boundary, stays in Ω, and converges to x_v. Along a trajectory, $T_v p = q$ reads $\frac{d}{dt} p(\Phi_t) = -\nabla v \cdot \nabla p(\Phi_t) = c_v p(\Phi_t) - q(\Phi_t)$, i.e. $\frac{d}{dt}\{J_t p(\Phi_t)\} = -J_t q(\Phi_t)$. Integrating over $t \in [0, \infty)$ gives (2.9) for every bounded solution, because $J_t p(\Phi_t) \to 0$ as $J_t \le e^{-c_0 t}$; this is uniqueness, and $\int_0^\infty e^{-c_0 t}\, dt = c_0^{-1}$ gives the supremum bound. Conversely, differentiating under the integral sign in (2.9) shows that it defines a smooth solution with bounded derivatives: this is legitimate because the operator norm of $D\Phi_t$ is at most e^{-mt} by (2.8), higher derivatives of Φ_t are bounded by polynomials in t, and every derivative of J_t is bounded by a polynomial in t times $J_t \le e^{-c_0 t}$. For the L^2 bound, Liouville's formula gives $\det D\Phi_t = \exp\{-\int_0^t \Delta v(\Phi_s)\, ds\} = J_t$, so the change of variables $y = \Phi_t(x)$ yields

$$\int_\Omega J_t(x)^2 |q(\Phi_t(x))|^2\, dx = \int_{\Phi_t(\Omega)} J_t(\Phi_t^{-1}(y)) |q(y)|^2\, dy \le e^{-c_0 t} \left\| q \right\|_2^2,$$

and Minkowski's integral inequality gives $\left\| p \right\|_2 \le \int_0^\infty e^{-c_0 t/2}\, dt \ \left\| q \right\|_2 = 2c_0^{-1} \left\| q \right\|_2$. By density, T_v^{-1} extends to $L^2(\Omega)$ with the same bound.

For the gradient, differentiate $T_v p = q$: $\nabla v \cdot \nabla(\nabla p) + (c_v I + D^2 v)\nabla p = \nabla q - p \nabla c_v$. This is a transport equation for the vector $w = \nabla p$ along the same flow, with the symmetric zero-order coefficient $\Lambda = c_v I + D^2 v \ge (c_0 + m) I$ and forcing $\Psi = \nabla q - p \nabla c_v$. With $w(t) = \nabla p(\Phi_t(x))$ we have $\dot w = \Lambda w - \Psi$, hence $\frac{d}{dt}|w|^2 = 2w \cdot \Lambda w - 2w \cdot \Psi \ge 2(c_0 + m)|w|^2 - 2|w||\Psi|$, so that $\frac{d}{dt}\{e^{-(c_0+m)t}|w(t)|\} \ge -e^{-(c_0+m)t}|\Psi(t)|$ wherever $w \ne 0$. Integrating from 0 to T and letting $T \to \infty$ (the term $e^{-(c_0+m)T}|w(T)|$ vanishes because w is bounded) gives $|\nabla p(x)| \le \int_0^\infty e^{-(c_0+m)t} |\Psi(\Phi_t(x))|\, dt$, which is (2.11). □

2.4 A variance gain from integration along characteristics

The forcing $\operatorname{div}(a\nabla e)$ involves second derivatives of the smoothed noise $e = n^{-1/2}S_h\mathbb{W}$, of pointwise variance $n^{-1}h^{-d-4}$. Integration along characteristics gains a factor h: the forcing is correlated over distance $2h$ while the flow moves at speed $|\nabla v| \asymp |x - x_v|$, so a trajectory spends time of order $h/|x - x_v|$ in a correlation ball, and $\int_{\Omega_h} \min\{1, h/|x - x_v|\}\, dx \asymp h$ for $d \geq 2$: the critical point costs nothing in L^2.

Lemma 2.4 (Variance gain)**.** *Let $0 < h < 1/2$, and let v be deterministic and satisfy the hypotheses of Lemma 2.3 on $\Omega = \Omega_h$. Let a be deterministic and classically continuously differentiable on $\overline{\Omega_h}$, with $\|a\|_{L^\infty(\Omega_h)} + \|\nabla a\|_{L^\infty(\Omega_h)} \leq A_0$. For $e = n^{-1/2}S_h\mathbb{W}$,*

$$\mathbb{E}\left\|T_v^{-1}\operatorname{div}(a\nabla e)\right\|^2_{L^2(\Omega_h)} \leq CA_0^2 n^{-1}h^{-d-3}, \tag{2.12}$$

where C depends only on d and the fixed kernel K.

Proof. Put $Z = \operatorname{div}(a\nabla e) = a\Delta e + \nabla a \cdot \nabla e$, a continuous Gaussian field on $\overline{\Omega_h}$; the flow formula (2.9) applies to it by uniform approximation with smooth forcings. Since $D^k e(z) = n^{-1/2}\mathbb{W}(D^k K_h(z - \cdot))$, the white-noise kernel of $Z(z)$ is $n^{-1/2}\{a(z)\Delta K_h(z - \cdot) + \nabla a(z) \cdot \nabla K_h(z - \cdot)\}$, a function supported in the ball of radius h around z with supremum at most $CA_0 n^{-1/2}h^{-d-2}$ (this uses the classical bound on a and ∇a). Hence, by scaling,

$$|\mathbb{E}Z(z)Z(z')| \leq CA_0^2 n^{-1}h^{-d-4}\mathbf{1}_{\{|z-z'|\leq 2h\}}. \tag{2.13}$$

Fix $x \in \Omega_h$ and let $R = |x - x_v|$. Since $\nabla v(y) = \int_0^1 D^2 v(x_v + \sigma(y - x_v))\, d\sigma\, (y - x_v)$, the Hessian bounds (2.8) give $e^{-Mt}R \leq |\Phi_t(x) - x_v| \leq e^{-mt}R$. For $s \geq 0$, applying the upper bound to the trajectory started at $\Phi_t(x)$ and the triangle inequality,

$$|\Phi_{t+s}(x) - \Phi_t(x)| \geq (1 - e^{-ms})|\Phi_t(x) - x_v| \geq (1 - e^{-ms})e^{-Mt}R.$$

The set of s for which the right-hand side is at most $2h$ has measure $O(he^{Mt}/R)$ within $[0, 1]$ and is empty in $[1, \infty)$ unless $he^{Mt}/R \geq c$; consequently

$$\int_0^\infty e^{-c_0 s}\mathbf{1}_{\{|\Phi_{t+s}(x)-\Phi_t(x)|\leq 2h\}}\, ds \leq C\min\{1, he^{Mt}/R\}.$$

Expanding the square of (2.9), using (2.13) and $J_t \leq e^{-c_0 t}$, and writing the double time integral over $\{t' \geq t\}$ with $t' = t + s$ (the other half is symmetric),

$$\mathbb{E}|T_v^{-1}Z(x)|^2 \leq CA_0^2 n^{-1}h^{-d-4}\int_0^\infty e^{-2c_0 t}\min\{1, he^{Mt}/R\}\, dt \leq CA_0^2 n^{-1}h^{-d-4}\min\{1, h/R\};$$

the last integral is evaluated by splitting at $he^{Mt} = R$ and uses $2c_0 = 1 > M = 3/(2d)$, valid for $d \geq 2$. (At $R = 0$ use the bound by a constant.) Finally, uniformly in $x_v \in \Omega_h$, $\int_{\Omega_h}\min\{1, h/|x - x_v|\}\, dx \leq Ch^d + Ch\int_h^2 \rho^{d-2}\, d\rho \leq Ch$ for $d \geq 2$, and integrating the pointwise bound over x proves the claim. □

3 Upper bound

3.1 The estimator

Take

$$h = n^{-1/(2\alpha+d+3)}, \qquad V_h = S_h Y_n \quad \text{on } \Omega_h, \tag{3.1}$$

and consider n large enough that $h < 1/2$. Here $V_h(x) = Y_n(K_h(x-\cdot))$ is the kernel smoother of the data, a smooth random function on a neighbourhood of $\overline{\Omega_h}$.

Gate and plug-in. If V_h satisfies the Hessian bounds $mI \le D^2 V_h \le MI$ of (2.8) on $\overline{\Omega_h}$ and $\nabla V_h \cdot \nu \ge 1/(4d)$ on $\partial\Omega_h$, set $\widetilde{f}_h = T_{V_h}^{-1} 1$ on Ω_h, given by Lemma 2.3 with $v = V_h$ and $q \equiv 1$; otherwise set $\widetilde{f}_h = 1$.

Boundary strip. We fill the strip $D \setminus \Omega_h$ by radial polynomial extrapolation, the reflection extension of Hestenes and Seeley [4, 10] adapted to a shrinking ball. Choose an integer $b > \alpha$ and constants $a_1, \dots, a_{b+1}$ with $\sum_{j=1}^{b+1} a_j (-j)^k = 1$ for $0 \le k \le b$ (a Vandermonde system). Define $E_h q = q$ on Ω_h, and for $0 < t \le h$, $\omega \in \mathbb{S}^{d-1}$,

$$E_h q((1-h+t)\omega) = \sum_{j=1}^{b+1} a_j q((1-h-jt)\omega). \tag{3.2}$$

For small h all radii on the right exceed $1/2$, so only values on Ω_h are used, and the moment identities say that E_h reproduces, along each ray, polynomials of degree at most b in the radial variable. A radial change of variables gives the first bound in

$$\|E_h q\|_{L^2(D)} \le C \|q\|_{L^2(\Omega_h)}, \qquad \|E_h(f|_{\Omega_h}) - f\|_{L^\infty(D)} \le C h^\alpha \|f\|_{C^\alpha(D)}. \tag{3.3}$$

For the second bound (at every index, integer or not), apply the extrapolation error to each dyadic block F_ℓ of a bounded extension F of f: Taylor's formula of order b and the moment identities bound $|E_h F_\ell - F_\ell|$ at distance t from $\partial\Omega_h$ by $C \min\{1, (t2^\ell)^{b+1}\} 2^{-\alpha\ell}$, and summing over ℓ gives Ct^α since $b + 1 > \alpha$.

Final estimator. Our estimator is $E_h \widetilde{f}_h$, clipped pointwise to $[1-r, 1+r]$; all operations involved are measurable. Every $f \in \mathcal{F}(\alpha, r)$ takes values in this interval and pointwise projection onto it is 1-Lipschitz, so clipping cannot increase the error, and the clipped estimator has loss bounded by a constant on any event.

3.2 The deterministic residual: bias cancellation

From now on $f \in \mathcal{F}(\alpha, r)$ is the true conductivity and $r \le r_0$ with r_0 as in Lemma 2.1, possibly decreased below. On Ω_h put $u_h = S_h u_f$, $f_h = S_h f$ and $e = V_h - u_h = n^{-1/2} S_h \mathbb{W}$: u_h, f_h are deterministic smoothed versions of the truth, and e is the smoothed noise, whose law does not depend on f. The deterministic u_h satisfies (2.8) with room to spare when r_0 is small: $S_h u_* = u_*$ by the kernel moments, so $u_h - u_* = S_h(u_f - u_*)$, and by (2.2)

$\sum_{j\le 2}\|D^j(u_h-u_*)\|_\infty \le Cr$; since $D^2u_* = d^{-1}I$ is the midpoint of $[m,M]$ and $\nabla u_*\cdot\nu = (1-h)/d \ge 1/(2d)$ on $\partial\Omega_h$, we may fix r_0 so that

$$\tfrac{3}{4}m\,I + \tfrac{1}{4}M\,I \le D^2u_h \le \tfrac{1}{4}m\,I + \tfrac{3}{4}M\,I, \qquad \nabla u_h\cdot\nu \ge \tfrac{3}{8d} \text{ on } \partial\Omega_h. \tag{3.4}$$

Moreover,

$$\sum_{j=0}^{1}\left\|D^j f_h\right\|_\infty + \sum_{j=0}^{2}\left\|D^j u_h\right\|_\infty \le C, \qquad \left\|D^2 f_h\right\|_\infty + \left\|D^3 u_h\right\|_\infty \le Ch^{-1}. \tag{3.5}$$

Indeed, by (2.2), f has a bounded gradient and u_f bounded second derivatives; convolution preserves these bounds, and placing one further derivative on K_h gives the second bound. In particular $\|f_h\|_\infty + \|\nabla f_h\|_\infty \le C$, which is the hypothesis on a in Lemma 2.4.

The key deterministic fact is that f_h almost solves the transport equation of u_h. On Ω_h, derivatives commute with S_h, so

$$f_h\nabla u_h = (S_hf)(S_h\nabla u_f), \qquad \operatorname{div} S_h(f\nabla u_f) = S_h\operatorname{div}(f\nabla u_f) = S_h1 = 1$$

by (1.1). Hence the residual is a divergence of a product-smoothing error, and Lemma 2.2 with $F = f$, $G = \partial_i u_f$ (both bounded in $C^\alpha(D)$ by (2.1)) and $k = 1, 2$ gives

$$B_h := T_{u_h}f_h - 1 = \operatorname{div}\{(S_hf)(S_h\nabla u_f) - S_h(f\nabla u_f)\}, \tag{3.6}$$

$$\|B_h\|_\infty + h\,\|\nabla B_h\|_\infty \le Ch^\alpha. \tag{3.7}$$

There is no boundary term because the convolution kernels used on Ω_h are supported inside D. This cancellation is why the bias is h^α rather than $h^{\alpha-1}$, the generic size of $D^2(u_h - u_f)$.

3.3 The noise event

For any fixed $A > 0$ there is an event $\mathcal{E}_n$ of probability at least $1 - C_An^{-A}$ on which, for $k = 0, 1, 2, 3$,

$$\left\|D^k e\right\|_{L^\infty(\Omega_h)} \le C_AL_nn^{-1/2}h^{-d/2-k} = C_AL_nh^{\alpha+3/2-k}, \tag{3.8}$$

the equality being the choice (3.1). This is a standard Gaussian supremum bound: $D^ke(x) = n^{-1/2}\mathbb{W}(D^kK_h(x-\cdot))$ has pointwise standard deviation at most $\sigma_k = Cn^{-1/2}h^{-d/2-k}$, and the canonical (covariance) distance between x and y is at most $\sigma_kC\min\{1, |x-y|/h\}$, since the two kernels differ by $O(|x-y|h^{-d-k-1})$ on a set of measure $O(h^d)$. Dudley's entropy bound and the Gaussian concentration inequality for suprema [3, Section 2.3 and Theorem 2.5.8] give (3.8), the factor L_n absorbing both $\sqrt{\log(1/h)}$ and the $\sqrt{A\log n}$ deviation.

On $\mathcal{E}_n$, for large n, the gate passes: by (3.4) and $\|D^2e\|_\infty + \|\nabla e\|_\infty \le CL_nh^{\alpha-1/2} \to 0$, the random $V_h = u_h + e$ satisfies (2.8) and $\nabla V_h\cdot\nu \ge 1/(4d)$. Moreover, by (3.5) and $\|D^3e\|_\infty \le CL_nh^{\alpha-3/2}$,

$$\left\|D^3V_h\right\|_\infty \le Ch^{-1} \quad \text{on } \mathcal{E}_n, \qquad \text{because } L_nh^{\alpha-1/2} \to 0. \tag{3.9}$$

3.4 Control of the plug-in error

Work on $\mathcal{E}_n$ with n large. Set $g = \widetilde{f}_h - f_h$ on Ω_h, the error of the plug-in relative to the smoothed truth. Since T_v is linear in v, $T_{V_h} = T_{u_h} + T_e$, and by (3.6), $T_{V_h} f_h = 1 + B_h + \operatorname{div}(f_h \nabla e)$; with $T_{V_h} \widetilde{f}_h = 1$, the exact equation for g is

$$T_{V_h} g = -B_h - \operatorname{div}(f_h \nabla e), \tag{3.10}$$

and since g is bounded, $g = T_{V_h}^{-1}$ of the right-hand side by uniqueness in Lemma 2.3 (the same remark applies to (3.12) below). By (3.7) and (3.5), the right-hand side is bounded by $C(h^\alpha + \|D^2 e\|_\infty + \|\nabla e\|_\infty)$ and its gradient by $C\{h^{\alpha-1} + \|D^3 e\|_\infty + \|D^2 e\|_\infty + h^{-1}\|\nabla e\|_\infty\}$, the last term coming from $D^2 f_h$. Applying the supremum bound (2.10) for $T_{V_h}^{-1}$, then (2.11) with $v = V_h$, where $\|\nabla c_{V_h}\|_\infty \le \sqrt{d}\,\|D^3 V_h\|_\infty \le C h^{-1}$ by (3.9), and inserting (3.8),

$$\|g\|_\infty \le C L_n h^{\alpha - 1/2}, \qquad \|\nabla g\|_\infty \le C L_n h^{\alpha - 3/2} \qquad \text{on } \mathcal{E}_n. \tag{3.11}$$

No independence between g and e is used.

3.5 Expansion in the deterministic operator and conclusion

Rewrite (3.10) with the deterministic transport operator on the left: $T_{u_h} g = T_{V_h} g - T_e g = -B_h - \operatorname{div}(f_h \nabla e) - \operatorname{div}(g \nabla e)$, so that on Ω_h

$$g = -T_{u_h}^{-1} B_h - T_{u_h}^{-1} \operatorname{div}(f_h \nabla e) - T_{u_h}^{-1} \operatorname{div}(g \nabla e). \tag{3.12}$$

The three terms are bias, linear noise and nonlinear remainder; the inverse is now deterministic, so Lemma 2.4 applies to the middle term. By (2.10) and (3.7), the first term has $L^2(\Omega_h)$ norm at most $C\|B_h\|_2 \le C h^\alpha$. Lemma 2.4, applied with the deterministic $v = u_h$ (which satisfies (2.8) by (3.4)) and $a = f_h$ (with $A_0 \le C$ by (3.5)), bounds the unconditional second moment of the second term by $C n^{-1} h^{-d-3}$. On $\mathcal{E}_n$, by (2.10), (3.11) and (3.8), the last term has $L^2(\Omega_h)$ norm at most

$$C\|\operatorname{div}(g \nabla e)\|_\infty \le C\big(\|g\|_\infty \big\|D^2 e\big\|_\infty + \|\nabla g\|_\infty \|\nabla e\|_\infty\big) \le C L_n^2 h^{2\alpha - 1}. \tag{3.13}$$

Finally, $\big\|E_h \widetilde{f}_h - f\big\|_{L^2(D)} \le C\big\|\widetilde{f}_h - f\big\|_{L^2(\Omega_h)} + C h^\alpha$ by (3.3), and $\big\|\widetilde{f}_h - f\big\|_{L^2(\Omega_h)} \le \|g\|_2 + \|f_h - f\|_{L^2(\Omega_h)} \le \|g\|_2 + C h^\alpha$ by (2.5). Using (3.12) on $\mathcal{E}_n$, the bounded loss of the clipped estimator on the complement, and clipping, the squared risk is bounded, uniformly in $f \in \mathcal{F}(\alpha, r)$, by

$$C\{h^{2\alpha} + n^{-1} h^{-d-3} + (\log n)^2 h^{4\alpha - 2} + n^{-A}\}. \tag{3.14}$$

With (3.1), the first two terms coincide and equal $n^{-2\alpha/(2\alpha+d+3)}$; since $\alpha > 1$, the third is $(\log n)^2 h^{2\alpha-2} \cdot h^{2\alpha} = o(h^{2\alpha})$, so the nonlinear remainder is negligible; and we may take $A = 2$. This proves the upper bound in Theorem 1.1.

4 Lower bound

4.1 Construction of the alternatives

We need conductivities in $\mathcal{F}(\alpha, r)$ that are well separated in $L^2(D)$ while their laws are close. Since $f \mapsto u_f$ is nonlinear, we prescribe exact solutions u_θ, so that the Kullback–Leibler divergences are exact Gaussian shifts, and recover $f_\theta = T_{u_\theta}^{-1} 1$ by Lemma 2.3; the only nonlinear quantity to control is then the deviation of $f_\theta - 1$ from its linearization $-T_{u_*}^{-1}\Delta(u_\theta - u_*)$.

In a fixed coordinate patch of the sphere $\mathbb{S}^{d-1}$ (of dimension $d-1 \geq 1$), place $N_h \asymp h^{-(d-1)}$ smooth bumps $\psi_{1,h}, \dots, \psi_{N_h,h}$ with pairwise disjoint supports of angular diameter proportional to h; they are translates and rescalings, in the chart, of one fixed nonzero bump. Uniformly in j,

$$\|\psi_{j,h}\|^2_{L^2(\mathbb{S}^{d-1})} \asymp h^{d-1}, \quad \|\Delta_{\mathbb{S}^{d-1}}\psi_{j,h}\|^2_{L^2(\mathbb{S}^{d-1})} \asymp h^{d-5}, \quad \left\|D^k \psi_{j,h}\right\|_\infty \leq C_k h^{-k}, \tag{4.1}$$

the derivatives being taken in the chart. All but the lower bound in the middle relation are scaling; for that one, integration by parts on the closed manifold $\mathbb{S}^{d-1}$ gives $\|\Delta\psi\|_2 \|\psi\|_2 \geq \|\nabla\psi\|_2^2$, and $\|\nabla\psi_{j,h}\|_2^2 \asymp h^{d-3}$ by scaling.

Choose a nonzero nonnegative radial profile $\eta \in C_c^\infty((1/4, 1/2))$. For $\theta \in \{0,1\}^{N_h}$ and a small fixed amplitude $\kappa > 0$, to be chosen in Section 4.2, prescribe the solutions

$$u_\theta = u_* + v_\theta, \qquad v_\theta(\rho\omega) = \kappa h^{\alpha+2}\eta(\rho)\sum_{j=1}^{N_h} \theta_j \psi_{j,h}(\omega). \tag{4.2}$$

The perturbation v_θ vanishes for $\rho < 1/4$ and for $\rho > 1/2$, so it is smooth on $\mathbb{R}^d$ and $u_\theta = 0$ on ∂D. The exponent $\alpha + 2$ is dictated by the parameter class: two angular derivatives of v_θ will become $f_\theta - 1$, which must have C^α norm of order κ. Since the bumps have disjoint supports, for every fixed integer $k \geq 0$,

$$\left\|D^k v_\theta\right\|_\infty \leq C_k \kappa h^{\alpha+2-k}, \qquad \|v_\theta\|_2^2 \leq C\kappa^2 h^{2\alpha+4}, \tag{4.3}$$

the second because $\sum_j \|\psi_{j,h}\|_2^2 \asymp N_h h^{d-1} \asymp 1$. In particular $\|D^2 v_\theta\|_\infty \leq C\kappa h^\alpha \leq C\kappa$, so for small κ the Hessians satisfy (2.8) on $\overline{D}$, and $\nabla u_\theta \cdot \nu = \nabla u_* \cdot \nu = 1/d > 0$ on ∂D. Lemma 2.3 applies on $\Omega = D$, and we define $f_\theta = T_{u_\theta}^{-1} 1 = 1 + q_\theta$. Since $T_{u_\theta} 1 = \Delta u_\theta = 1 + \Delta v_\theta$, the deviation q_θ solves the transport equation

$$T_{u_\theta} q_\theta = -\Delta v_\theta. \tag{4.4}$$

4.2 Membership in the parameter class

We claim that, for every fixed integer $k \geq 0$ (only $k \leq \lfloor \alpha \rfloor + 1$ is needed),

$$\left\|D^k q_\theta\right\|_\infty \leq C_k \kappa h^{\alpha-k}. \tag{4.5}$$

For $k = 0$, (4.4), the supremum bound (2.10) and (4.3) give $\|q_\theta\|_\infty \le c_0^{-1} \|\Delta v_\theta\|_\infty \le C\kappa h^\alpha$. For the induction step, differentiate (4.4) k times. The result is a transport equation for the symmetric tensor $D^k q_\theta$ along the same flow, in which the first-order part is $\nabla u_\theta \cdot \nabla(D^k q_\theta)$ and the zero-order principal coefficient is Δu_θ plus $D^2 u_\theta$ acting in each of the k tensor slots (one such term for each way of placing exactly one derivative on ∇u_θ); this symmetric coefficient is bounded below by $c_0 + km$. The main forcing is $D^k \Delta v_\theta$, of size $C_k \kappa h^{\alpha-k}$. The remaining terms come from placing at least two derivatives on ∇u_θ, or at least one on Δu_θ; since u_* is quadratic, only v_θ contributes, and they are products of the forms

$$D^j \Delta v_\theta \, D^{k-j} q_\theta \quad (1 \le j \le k), \qquad D^{j+1} v_\theta \, D^{k+1-j} q_\theta \quad (2 \le j \le k).$$

By (4.3) and the induction hypothesis (which applies since $k-j \le k-1$ and $k+1-j \le k-1$), each is bounded by $C_k \kappa^2 h^{2\alpha-k} \le C_k \kappa h^{\alpha-k}$. The tensor version of the flow bound, proved exactly as (2.11) with the Frobenius norm of the tensor in place of $|w|$, gives (4.5).

From (4.5) to the Hölder–Zygmund norm: if α is not an integer, the supremum bound for $D^{\lfloor\alpha\rfloor} q_\theta$ and the Lipschitz bound coming from $D^{\lfloor\alpha\rfloor+1} q_\theta$ interpolate to the Hölder seminorm of exponent $\alpha - \lfloor\alpha\rfloor$, with the powers of h cancelling exactly; if α is an integer, bounded classical derivatives of order α control the Zygmund norm. (These classical bounds on $\overline{D}$ pass to the quotient norm of Section 1.1 through a bounded extension to $\mathbb{R}^d$, e.g. the reflection extension [4, 10] across the smooth boundary.) Thus

$$\|f_\theta - 1\|_{C^\alpha(D)} \le C\kappa. \tag{4.6}$$

Fix $\kappa > 0$ so small that the right-hand side is at most r and the Hessian condition above holds. Then all f_θ belong to $\mathcal{F}(\alpha, r)$, in particular they are positive. By construction $\operatorname{div}(f_\theta \nabla u_\theta) = 1$ in D and $u_\theta|_{\partial D} = 0$, so uniqueness of the weak solution gives $u_\theta = u_{f_\theta}$ exactly.

4.3 Separation of the alternatives

Let $T_* = T_{u_*} = d^{-1} x \cdot \nabla + 1$ be the transport operator of the reference solution, and let $w_\theta = -T_*^{-1} \Delta v_\theta$ be the linearization of q_θ. Since $T_{u_\theta} = T_* + T_{v_\theta}$, (4.4) gives the exact identity $T_* q_\theta = -\Delta v_\theta - \operatorname{div}(q_\theta \nabla v_\theta)$, so $q_\theta - w_\theta = -T_*^{-1} \operatorname{div}(q_\theta \nabla v_\theta)$, and (2.10), (4.3) and (4.5) imply

$$\|q_\theta - w_\theta\|_2 \le C \, \|\operatorname{div}(q_\theta \nabla v_\theta)\|_\infty \le C\big(\|\nabla q_\theta\|_\infty \, \|\nabla v_\theta\|_\infty + \|q_\theta\|_\infty \, \|\Delta v_\theta\|_\infty \big) \le C\kappa^2 h^{2\alpha}. \tag{4.7}$$

The inverse of T_* acts along rays: solving $d^{-1}\rho \, \partial_\rho w + w = z$, i.e. $\partial_\rho(\rho^d w) = d\rho^{d-1} z$, gives $T_*^{-1} z(\rho\omega) = d\rho^{-d} \int_0^\rho s^{d-1} z(s\omega) \, ds$. In polar coordinates $\Delta(\eta\psi) = (\eta'' + (d-1)\rho^{-1}\eta')\psi + \rho^{-2}\eta \, \Delta_{\mathbb{S}^{d-1}}\psi$, and the radial part integrates to $\int_0^\rho s^{d-1}(\eta'' + (d-1)s^{-1}\eta') \, ds = \rho^{d-1}\eta'(\rho)$, which vanishes for $\rho > 1/2$, outside the support of η. Therefore, for $1/2 < \rho < 1$,

$$w_\theta(\rho\omega) = -d\kappa h^{\alpha+2} \rho^{-d} A_\eta \Delta_{\mathbb{S}^{d-1}} \sum_j \theta_j \psi_{j,h}(\omega), \qquad A_\eta = \int_{1/4}^{1/2} s^{d-3} \eta(s) \, ds > 0 : \tag{4.8}$$

the two angular derivatives convert the amplitude $h^{\alpha+2}$ into h^{α}. If the Hamming distance between θ, θ' is at least cN_h, the disjointness of the supports of the $\Delta_{\mathbb{S}^{d-1}}\psi_{j,h}$, (4.1) and $N_h h^{d-1} \asymp 1$ give $\|w_\theta - w_{\theta'}\|_2^2 \geq c\kappa^2 h^{2\alpha+4} \cdot cN_h\, h^{d-5} \geq c'\kappa^2 h^{2\alpha}$. By (4.7) and the triangle inequality, for all sufficiently small h the corresponding conductivities satisfy

$$\|f_\theta - f_{\theta'}\|_{L^2(D)} = \|q_\theta - q_{\theta'}\|_2 \geq c''\kappa h^{\alpha}. \tag{4.9}$$

4.4 Information bound and Fano's inequality

The Varshamov–Gilbert bound [12, Lemma 2.9] provides a subset $\Theta_h \subset \{0,1\}^{N_h}$ with pairwise Hamming distances at least $N_h/8$ and $\log|\Theta_h| \geq cN_h \geq ch^{-(d-1)}$. For the Gaussian white noise model (1.2), the Kullback–Leibler divergence between two shifts is the exact Gaussian expression [3, equation (6.16)], so, writing $P_{\mathbf{1}}$ for the law of Y_n under $f = \mathbf{1}$, (4.3) gives

$$\mathrm{KL}(P_{f_\theta}, P_{\mathbf{1}}) = \frac{n}{2}\,\|u_\theta - u_*\|_2^2 = \frac{n}{2}\,\|v_\theta\|_2^2 \leq Cn\kappa^2 h^{2\alpha+4}. \tag{4.10}$$

Choose $h = c_* n^{-1/(2\alpha+d+3)}$ with $c_* > 0$ small. Then $nh^{2\alpha+4} = c_*^{2\alpha+d+3} h^{-(d-1)}$, so (4.10) is at most $\frac{1}{16}\log|\Theta_h|$ for small c_*.

Let ϑ be uniformly distributed on Θ_h and $\bar{P}$ the mixture of the P_{f_θ}. For any probability measure Q with respect to which the P_{f_θ} have finite Kullback–Leibler divergence, writing $dP_{f_\theta}/d\bar{P}$ as $(dP_{f_\theta}/dQ)/(d\bar{P}/dQ)$ inside the logarithm and averaging over θ shows that the mutual information between ϑ and Y_n satisfies

$$\begin{aligned} I(\vartheta; Y_n) &= |\Theta_h|^{-1} \textstyle\sum_\theta \mathrm{KL}(P_{f_\theta}, \bar{P}) = |\Theta_h|^{-1} \textstyle\sum_\theta \mathrm{KL}(P_{f_\theta}, Q) - \mathrm{KL}(\bar{P}, Q) \\ &\leq |\Theta_h|^{-1} \textstyle\sum_\theta \mathrm{KL}(P_{f_\theta}, Q) \leq \tfrac{1}{16}\log|\Theta_h| \qquad \text{for } Q = P_{\mathbf{1}}. \end{aligned}$$

Fano's inequality [12, Lemma 2.10 and Corollary 2.6] therefore bounds the average misclassification probability of every classifier $\widehat{\vartheta} = \widehat{\vartheta}(Y_n)$ with values in Θ_h below by

$$\frac{1}{|\Theta_h|}\sum_{\theta\in\Theta_h} P_{f_\theta}(\widehat{\vartheta} \neq \theta) \;\geq\; 1 - \frac{\frac{1}{16}\log|\Theta_h| + \log 2}{\log|\Theta_h|} \geq \frac{1}{2}$$

for large n. Given an arbitrary estimator $\widehat{f}_n$, let $\widehat{\vartheta}$ be the index of a nearest f_θ in $L^2(D)$. By (4.9), $\widehat{\vartheta} \neq \theta$ forces $\left\|\widehat{f}_n - f_\theta\right\|_2 \geq c''\kappa h^{\alpha}/2$, so

$$\max_{\theta\in\Theta_h} \mathbb{E}_{f_\theta}\left\|\widehat{f}_n - f_\theta\right\|_2^2 \geq \frac{(c''\kappa h^{\alpha})^2}{4}\cdot\frac{1}{|\Theta_h|}\sum_{\theta} P_{f_\theta}(\widehat{\vartheta} \neq \theta) \geq c\kappa^2 h^{2\alpha} = c\, n^{-\frac{2\alpha}{2\alpha+d+3}}.$$

Since all f_θ lie in $\mathcal{F}(\alpha, r)$, this is the claimed lower bound and completes the proof of Theorem 1.1.

5 Remarks on the mechanism

Remark 5.1. *The two halves of the proof explain the denominator* $2\alpha+d+3 = 2\alpha+(d-1)+4$ *from opposite directions: the alternatives oscillate in the* $d-1$ *angular directions and their solutions are smaller than their conductivities by two angular derivatives, while integration along the gradient flow averages the noise in the one flow direction but not in the* $d-1$ *transverse ones. No boundary values of* f *were assumed, and the linearized alternatives* (4.8) *are nonconstant up to the boundary. Besides the classical embedding,* $\alpha > 1$ *enters through* $\sum_i 2^{(1-\alpha)i} < \infty$ *and* $2\alpha > \alpha + 1$ *in the proof of Lemma 2.2, and through the squared nonlinear remainder* (3.13) *being* $o(h^{2\alpha})$.

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